THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE FULL SYLLABUS PAPER SET 03

Full Syllabus (Chapters 1–13) | RBSE Pattern | Session 2026–27

Time: 3 Hours 15 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. To model a cricket shot that may go for a six, which detail is MOST important? 1

(a) the amount of grass on the field
(b) the colour of the ball
(c) the mass of the ball and its speed and direction
(d) the brand of the bat

2. Which of these is a good example of a scientific prediction? 1

(a) Guessing the winner by colour of jersey
(b) Saying the match will be lucky
(c) Choosing numbers at random
(d) Using the speed and angle of a kick to find how far a football will travel

3. A mixture of camphor and sand can be separated by: 1

(a) separating funnel **(b)** sublimation
(c) crystallization **(d)** distillation

4. The pure liquid obtained by condensing vapour in distillation is called the: 1

(a) filtrate **(b)** distillate **(c)** residue **(d)** precipitate

5. If 2.4 g of carbon reacts completely with oxygen (C : O = 12 : 32), the mass of CO_2 formed is: 1

(a) 4.4 g **(b)** 10.4 g **(c)** 8.8 g **(d)** 6.2 g

6. The chemical formula of calcium carbonate is: 1

(a) CaC_2O_3 **(b)** $CaCO_3$ **(c)** $Ca(CO_3)_2$ **(d)** Ca_2CO_3

7. An atom of aluminium has atomic number 13 and mass number 27. The number of neutrons in it is: 1

(a) 15 **(b)** 13 **(c)** 14 **(d)** 27

8. Which equation is NOT a kinematic equation for constant acceleration? 1

(a) $v^2 = u^2 + 2as$ **(b)** $v = u + at$ **(c)** $v = s/t^2$ **(d)** $s = ut + \frac{1}{2}at^2$

9. A horizontal line on a velocity–time graph shows: 1

(a) rest **(b)** decreasing velocity
(c) constant acceleration **(d)** constant velocity, zero acceleration

10. Two forces of 10 N and 6 N act on a block along the same straight line in opposite directions. The magnitude of the net force is: 1

(a) 60 N **(b)** 16 N **(c)** 4 N **(d)** 10 N

11. Grooves on shoe soles and tyre treads are made to: 1

(a) reduce weight **(b)** increase friction
(c) increase speed **(d)** reduce friction

12. The kinetic energy of a body of mass 1000 kg moving at 10 m s^{-1} is: 1

(a) 25000 J **(b)** 100000 J **(c)** 10000 J **(d)** 50000 J

13. Thunder is heard 5 s after lightning is seen. Taking v = 340 m s^{-1}, the lightning is at a distance of: 1

(a) 3400 m **(b)** 1700 m **(c)** 850 m **(d)** 1710 m

14. If a cell is 200 µm in size, in millimetres it is: 1

(a) 0.2 mm **(b)** 2 mm **(c)** 20 mm **(d)** 0.02 mm

15. Prokaryotic cells differ from eukaryotic cells in not having: 1

(a) cytoplasm **(b)** a well-defined nucleus
(c) DNA **(d)** cell membrane

16. Food prepared in leaves is transported to other parts by: 1

(a) cork **(b)** phloem **(c)** xylem **(d)** stomata

17. Fungal cell walls are made up of: 1

(a) chitin **(b)** peptidoglycan only
(c) lignin **(d)** cellulose

18. After fertilisation, the ovary develops into a: 1

(a) fruit **(b)** flower **(c)** stamen **(d)** seed

19. Spore formation is seen in: 1

(a) hydra **(b)** bread mould (Rhizopus)
(c) human **(d)** potato

20. The speed of light in vacuum is: 1

(a) 3×10^{8} m s^{-1} **(b)** 3×10^{10} m s^{-1} **(c)** 3×10^{6} m s^{-1} **(d)** 340 m s^{-1}

SECTION — B (Very Short Answer Type) 8 × 2 = 16

Answer in about 10–15 words each.

21. Give two questions that would improve the prediction ‘it will rain this afternoon because clouds look dark’. 2

22. Differentiate between molecular mass and formula unit mass. 2

23. Give one example each of uniform and non-uniform motion in a straight line. 2

24. A box is pushed with a force equal to the friction acting on it. Will it move with constant velocity? Why? 2

25. Name three types of plastids and their functions. 2

26. State the function of the rib cage. 2

27. Why must gametes have half the number of chromosomes? 2

28. What is the ozone layer? Why is it important? 2

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SECTION — C (Short Answer Type) — 8 × 3 = 24

Answer in about 30–40 words each.

29. Why are most gas–gas mixtures homogeneous? Give one example and one example of a heterogeneous mixture of a gas. 3

30. Two atoms have 11 protons each, one with 12 and the other with 13 neutrons. Compare their atomic numbers and mass numbers. Are they the same element? 3

31. Why can a perpetual-motion machine not exist? 3

32. What is noise pollution? State two harmful effects and two measures. 3

33. What is the role of nervous tissue? Name its structural unit. 3

34. Why are gymnosperms better adapted to dry conditions than pteridophytes? 3

35. Write the formulae of 3
(i) aluminium oxide
(ii) calcium carbonate
(iii) sodium sulfate using the criss-cross method for one of them.

OR

Why do ionic compounds have high melting points and are usually hard but brittle? 3

36. How do planetary winds and pressure belts form? 3

OR

Name three factors that drive ocean currents. 3

SECTION — D (Long Answer Type) — 4 × 5 = 20

Answer in about 100–150 words each.

37. (i) What is nuclear energy? Name one use. 5
(ii) Name the scientist called the Father of the Indian Nuclear Programme.
(iii) How does a nuclear power plant produce electricity?

OR

Case study: A chemist records the substances CO_2, NH_3, NaCl and $MgCl_2$. 5
(i) Classify each as molecular (covalent) or ionic.
(ii) Find the molecular mass of CO_2 and NH_3.
(iii) Find the formula unit mass of NaCl and $MgCl_2$.
(iv) Why are formula unit masses used for ionic compounds? (H=1, C=12, N=14, O=16, Na=23, Mg=24, Cl=35.5)

38. (i) State the three laws with an example each. 5
(ii) Which law explains why seat belts are essential?

OR

(i) Explain the system approach for two boxes of m_1 and m_2 pulled by force F. 5
(ii) Find the acceleration for m_1 = 2 kg, m_2 = 3 kg, F = 10 N.
(iii) Find the tension.

39. (i) Describe plastids and their types. 5
(ii) State two features that resemble bacteria.
(iii) Rohit says roots do not have plastids. Is he correct?

OR

Case study: Onion root tip cells show different shapes of chromatin under the microscope. 5

(i) Why are root tips used?

(ii) What process is seen?

(iii) Name its types.

(iv) What is the use of this process in the body?

40. (i) What is the role of meiosis in sexual reproduction? 5

(ii) How many chromosomes does a human sperm, egg and zygote have?

(iii) Explain how variation arises among siblings.

OR

(i) Describe the development of the foetus in the three trimesters. 5

(ii) Role of the placenta.

(iii) Name two things a mother should avoid in pregnancy.

END OF QUESTION PAPER **Total: 80 Marks**

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